

PROJECT ROLES & RESOURCE ALLOCATION

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Project Developer Profile

The OptiCrop project was developed entirely as an individual project. The developer executed all roles across the software development lifecycle:

Development Role	Responsibilities Executed	Key Deliverables Completed
Product Lead	Requirements elicitation, planning, and documentation.	SRs, empathy mapping, and user journey maps.
ML Engineer	Data preprocessing, algorithm selection, and training.	Trained model pipelines and evaluation plots.
Backend Developer	Flask server controller logic, route validation, database logic.	api.py, routes, predictions.csv, health check routes.
UI/UX Designer	Glassmorphic dashboard style guidelines, responsive layout.	index.html, result.html, dashboard.html templates.

- Member A - Lead Presenter

Member A - Lead Presenter

Delivers introduction, problem framing, closing roadmap & Q&A.
- Member B - ML Presenter

Member B - ML Presenter

Explains dataset, model selection, accuracy metrics, confusion matrix.
- Member C - App Presenter

Member C - App Presenter

Performs live web-app demo: form input -> crop result -> dashboard.
- Member D - QA Presenter

Member D - QA Presenter

Runs pytest live, shows CSV log, explains test coverage.
- Member D - QA Presenter

Member D - QA Presenter

Runs pytest live, shows CSV log, explains test coverage.